Supplementary Material

Two-Time-Scale Coordination of Vehicle-to-Grid Voltage Support and Frequency Reserve in a Two-Area System Using Proximal Policy Optimization

S1. Reproducibility

All results are generated by the released code from the raw simulation records. The repository (`https://github.com/kenoto99/v2g-ppo-two-area`, release v3.0.1; Zenodo DOI 10.5281/zenodo.23851585) contains `REPRODUCE.md` with the ordered commands, `results/ENVIRONMENT.txt` with the exact package versions, `results/CHECKSUMS.sha256` with SHA-256 checksums of every raw result, table, figure, trained policy and PDF, and `results/verification_report.md`, the output of `scripts/verify_all.py`. That script re-derives every number in the manuscript independently of the script that builds the tables and compares the two; it also re-runs the pandapower verification and the physics tests and checks that every trained policy carries the final model version tag (`env_version = v3`).

Random seeds. PPO training seeds 0 to 4 (main), 0 to 2 (ablation, squashed policy), 0 and 1 (weight sensitivity). Training-day generator `default_rng(1000 seed + env)` for four parallel environments; validation days `default_rng(10000)`; test days `default_rng(2026)`; stress days `default_rng(777)`; the disturbance of each hour `default_rng([scenario seed, hour])`, identical for every controller and treatment.

Package versions. Python 3.13, NumPy 2.4.6, SciPy 1.18.1, PyTorch 2.15.0 (CUDA build, executed on CPU), Gymnasium 1.4.0, Stable-Baselines3 2.9.2, pandapower 3.5.6, numba 0.68.1, matplotlib 3.11.2.

S2. Learning curves of all runs

Table 1 lists every training run with its final validation cost, the change of the validation cost over the last 50 000 steps (convergence criterion 2%), and wall-clock time on one CPU core.

Table 1: All training runs (final model version v3).

Run	Steps	Reward config	Final val. cost	Best val. cost	Change, last 50k (%)	Wall (min)
ppo_main_s0	400k	nominal	136.8	136.6	0.1	27
ppo_main_s1	400k	nominal	138.3	138.3	1.0	27
ppo_main_s2	400k	nominal	136.8	136.8	0.8	36
ppo_main_s3	400k	nominal	136.5	136.4	0.0	36
ppo_main_s4	400k	nominal	140.2	139.6	0.2	27
ppo_noDeg_s0	200k	'w_deg': 0.0	137.5	135.8	0.4	12
ppo_noDeg_s1	200k	'w_deg': 0.0	155.4	155.4	5.3	12
ppo_noDeg_s2	200k	'w_deg': 0.0	138.2	137.1	3.1	12
ppo_noF_s0	200k	'w_f': 0.0, 'w_tie': 0.0	81.0	80.2	0.4	13
ppo_noF_s1	200k	'w_f': 0.0, 'w_tie': 0.0	81.3	81.3	0.7	14
ppo_noF_s2	200k	'w_f': 0.0, 'w_tie': 0.0	81.8	80.0	0.8	12
ppo_noOV_s0	200k	'w_ov': 0.0	143.4	142.8	7.8	29
ppo_noOV_s1	200k	'w_ov': 0.0	153.8	153.8	9.0	22
ppo_noOV_s2	200k	'w_ov': 0.0	144.3	143.8	11.4	14
ppo_ref200_s0	200k	nominal	143.3	142.1	4.9	25
ppo_ref200_s1	200k	nominal	161.9	161.9	4.0	29
ppo_ref200_s2	200k	nominal	144.7	144.2	10.8	23
ppo_squash_s0	200k	nominal, squashed	178.2	177.7	0.2	15
ppo_squash_s1	200k	nominal, squashed	178.5	178.2	0.4	14
ppo_squash_s2	200k	nominal, squashed	177.9	177.5	0.0	27
ppo_wF2_s0	200k	'w_f': 2.0	119.3	119.3	4.8	19
ppo_wF2_s1	200k	'w_f': 2.0	126.2	126.2	0.7	12
ppo_wF8_s0	200k	'w_f': 8.0	219.6	218.3	0.4	12
ppo_wF8_s1	200k	'w_f': 8.0	217.7	217.7	4.3	12
ppo_wV100_s0	200k	'w_v': 100.0	213.5	213.0	0.0	12
ppo_wV100_s1	200k	'w_v': 100.0	213.2	213.2	1.7	19
ppo_wV25_s0	200k	'w_v': 25.0	114.4	114.2	0.1	12
ppo_wV25_s1	200k	'w_v': 25.0	113.2	113.0	1.8	12

S3. Model verification details

Base case of the IEEE 33-bus feeder: minimum voltage 0.9131 p.u. at bus 18, loss 202.68 kW. Largest voltage difference from pandapower over 20 random operating points: 1.2×10^{-10} p.u. Two-area LFC without AGC after a 0.02-p.u. step: $\Delta f = -0.02381$ Hz (theory -0.02381), $\Delta P_{\text{tie}} = -5.000$ MW (theory -5.000). The physics tests in `tests/test_physics.py` additionally verify the regulation-energy integral against a closed-form case, the feasibility of random (including out-of-range) actions on 40 train and stress days with the 60-s and the exact 3600-s fast layer, that no upward reserve is credited when driving alone takes the fleet below its floor, and that raw-action clipping and physical projection are counted separately.

S4. Regulation-energy treatments, per controller

Table 5 of the main text gives the daily results under each treatment. The tracking statistics against the exact 3600-s simulation on the test days are: gross energy ratio (treatment / exact) 2.9 (hourly extrapolation), 0.94 (decomposed) for PPO; 2.7 and 0.94 for RB; 2.6 and 0.94 for DAO. Net daily energy (MWh): PPO -0.73 exact, -0.69 decomposed, -1.67 extrapolated, -0.03 window; RB -0.50, -0.46, -1.10, -0.02; DAO -0.47, -0.43, -1.03, -0.02. Share of hours in which the extrapolation has the wrong sign: PPO 39%, RB 43%, DAO 43%. Fraction of simulated time at the droop saturation limit in the exact simulations: PPO 0.00%, RB 0.00%, DAO 0.00%. Minimum sustained-activation margin over test and stress days: PPO 0.000 MWh, RB 0.701 MWh, DAO 0.000 MWh.

S5. Raw-action saturation of the final policies

Share of hourly decisions above $+1$ / below -1 before clipping (test days, five seeds): a_1 13.4% / 25.1%, a_2 21.6% / 33.9%, a_3 43.2% / 10.8%, a_4 30.7% / 17.9%. Median excess beyond the bound when saturated: 0.61.

S6. Analyses added after the technical review of the revised manuscript

Three analyses were added in response to the technical review of the revised manuscript. They reuse the trained policies and the test-day scenarios of the main evaluation and add no new training run.

- `scripts/dao_rho_sweep.py` re-optimizes the day-ahead benchmark for reserve shares $\rho \in \{0.0, 0.1, \ldots, 0.6\}$ on the test and stress days and reports a per-day oracle (Table 4 of the manuscript). The $\rho = 0.3$ row reproduces the DAO column of Table 2.
- `scripts/activation_powerflow.py` re-solves every test-day hour at $P^{\text{set}} \pm R$ and at the realized peak droop power of the simulated window and records voltages and branch currents (Table 8). The schedule case reproduces the violation hours of Table 2.
- `scripts/chained_days.py` runs the 30 test days back to back with the terminal fleet energy carried into the next day (Table 3). Day 1 reproduces the first independent test day exactly.

The verification script checks each of these consistency conditions and recomputes the corresponding macros.

S7. Remaining model assumptions

Each area is represented by 25 identical copies of one IEEE 33-bus feeder; the fleet of each feeder is one aggregated battery with a single SoC, so SoC dispersion and the reduced deliverable power of vehicles at their limits are not represented; charger reactive power is not used; chargers are a first-order lag with saturation, without current limits or control-loop dynamics; communication delays between the aggregator and the chargers are not modeled; all chargers and vehicles are identical; LFC and fleet parameters are fixed at representative values and were not varied in training or testing; the LFC model is linear with representative parameters; load, PV and availability profiles are synthetic and shaped after published measurements; hour-ahead forecasts are assumed exact; the terminal SoC requirement is a soft penalty, and consecutive-day operation shows a downward drift of the residential fleet energy for every controller (Table 3); the fast layer of each hour is a 60-s window containing one step event and a fluctuating component, and the regulation energy of the remaining 59 minutes is obtained from a second fluctuation-only window (checked against exact 3600-s simulations); the throughput term is a linear proxy for degradation; the weights of the reward were chosen by the authors and their effect is shown in the ablation and sensitivity studies.